KERALA SSLC | MATHEMATICS

POLYNOMIALS AND EQUATIONS

PYQ • 2021–2026

1. 2026 **4 MARKS**

Write the polynomial $x^2 - 12x + 32$ as the product of two first-degree polynomials. **(3)**

For what values of x does the polynomial $x^2 - 12x + 32$ become zero? **(1)**

2. 2025 **4 MARKS**

The length of a rectangular hall is 5 metres more than the breadth. The diagonal is 10 metres more than the breadth.

(a) Taking breadth as x, write the measures of length and diagonal in terms of x.

(b) Form a second-degree equation and find the length and breadth of the hall.

3. 2024 **4 MARKS**

Consider the polynomial $p(x) = x^2 - 5x + 6$.

(a) Write p(x) as the product of two first-degree polynomials.

(b) Find the solutions of the equation p(x) = 0.

4. 2023 **4 MARKS**

$P(x) = x^2 - 4x + 4$.

(a) What is P(1)? **(1)**

(b) Write a first-degree factor of P(x) – P(1). **(1)**

(c) Write the polynomial P(x) – P(1) as the product of two first-degree polynomials. **(2)**

5. 2022 **2 MARKS**

Write the second-degree polynomial $x^2 + x$ as the product of two first-degree polynomials.

6. 2022 **1 MARK**

If $(x - 1)$ is a factor of the polynomial $p(x)$, write $p(1)$.

7. 2021 **2 MARKS**

Write $x^2 - 1$ as the product of two first-degree polynomials.

8. 2021 **4 MARKS**

$P(x) = x^2 - 5x + 9$.

(a) Find $P(2)$ and $P(3)$. **(2)**

(b) Write $P(x) - P(2)$ as the product of two first-degree polynomials. **(2)**